STA 141C Final Project
Group: Bao, Rathor, Schneider, Soto

Death and Survival relating to Heart Attacks within Hospital Walls (Myocardial Infarction Prediction)

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1. Introduction

1.1. Background

Heart disease remains a significant global health issue, with myocardial infarction (MI) being one of the most prevalent and deadly forms. According to the American Heart Association and CDC, heart disease is the leading cause of death in the United States, accounting for nearly one in every four deaths. The economic impact is equally profound, with billions of dollars spent annually on healthcare costs and lost productivity. Given the severity and prevalence of MI, there is a critical need for effective predictive models that can identify patients at high risk of adverse outcomes, thereby enabling timely and targeted interventions.

1.2. Proposed Question

This project aims to address the question: Can we develop a method to identify a patient's risk of heart-related death? By leveraging a vast dataset and pattern recognition techniques, our goal is to build a predictive model that can accurately foresee a patient's potential lethal outcome based on associated factors.

1.3. Our Dataset

The dataset utilized in this study is the "Myocardial Infarction Complications" dataset, sourced from the UCI Machine Learning Repository. The value of this dataset lies in its comprehensive range of 111 features, which enhances our ability to predict heart attacks. With such a diverse set of features, we can identify key predictors and explore correlations that may influence our target variable. This dataset contains 1700 patient records, which allows for the development of more accurate and reliable models. The features include 22 categorical variables, 79 binary variables, and 11 continuous variables, providing a rich set of predictors for our analysis.

The original dataset's target feature was categorical, with 0 indicating patient survival and values 1 through 7 categorizing different types of heart attack deaths. Since our proposed question focuses solely on whether the patient survives or dies, we combined the categories 1 through 7 into a single category. Thus, the target variable was transformed into a binary indicator, serving as the primary endpoint for our predictive model. One of the significant challenges with this dataset is the presence of 15,974 missing values, requiring data imputation to maintain the integrity and reliability of our analysis.

2. Exploratory Data Analysis

2.1. Preprocessing

The first step was to address the missing values, which constituted a substantial portion of the dataset. Rows and columns with more than 25% missing data were immediately removed. This initial reduction step ensured that the subsequent imputation process would be more manageable and accurate.

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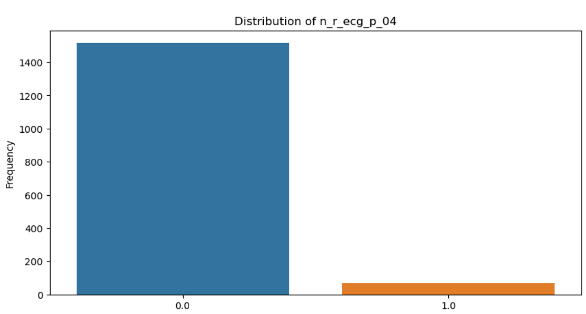
Next, we employed two imputation methods: K-Nearest Neighbors (KNN) and Iterative Imputer with Random Forest. KNN imputation is a non-parametric method that estimates missing values based on the values of the nearest neighbors. It does so by first identifying similar samples using the available parameters not missing in the sample of interest, and comparing them with those same parameters across all samples in the dataset. Then, using a distance measurement (a common measurement being Euclidean distance) it then groups the sample of interest with the "nearest neighbors," or those with the least distance from itself. If the parameter of interest is continuous, a common estimation technique is to average the same parameter values across the nearest neighbors and impute such into the missing sample. Should the parameter be categorical, the algorithm would select the most common category. This method is computationally efficient but tends to not perform well for datasets with high dimensionality or mixed data types. This method struggles with binary variables in particular, as common distance measurements do not work well with binary data.

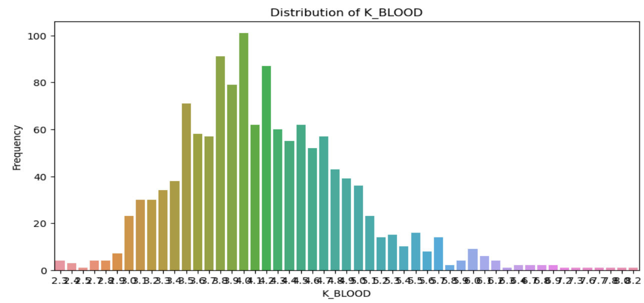
In contrast, the Iterative Imputer with Random Forest is a more sophisticated technique that iteratively models each feature as a function of the others, allowing for greater flexibility and accuracy, especially for mixed data types. This method works by first imputing all missing values with a placeholder value. For continuous data types, this can just be the average of all values within that parameter. For categorical or binary data, this could take the form as the most occurring category within the total parameter. Following this, the model would then examine each newly imputed data value within the parameter one at a time, using all other parameters as predictor variables for the target parameter via the random forest method. Once this has been completed, and the new values have been imputed (overwriting the initial placeholder values), the model then moves on to the next column and repeats the process. This cycle will continue until all parameters have been examined and updated. Following this, the process would be run again and again until the imputed values begin to stabilize and see negligible change with each iteration.

As mentioned earlier, the Iterative Imputer with Random Forest method is computationally more strenuous and difficult to utilize when examining large datasets. We navigated around this pitfall by parallelizing our code, allowing it to run much faster. Since the imputed data using the Iterative Imputer method was more reliable across all data types, we opted to use this method to bolster the accuracy of our findings.

2.2. Visualization

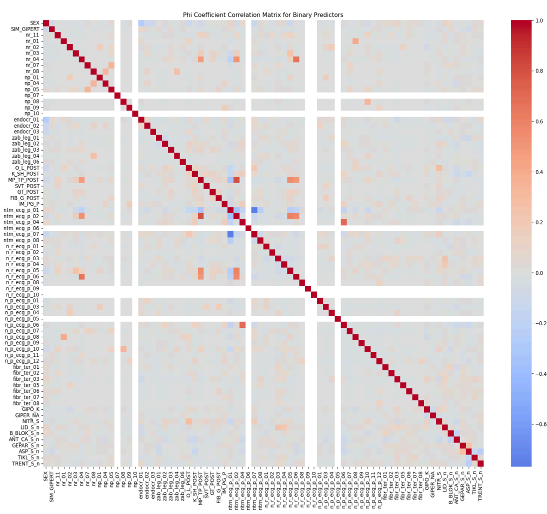


To better understand the dataset, it was necessary to visualize the trends, patterns and correlations. Through exploring the binary variables using boxplots, as mentioned above that far more patients in the dataset survived, it meant that many of the medical conditions that lead to fatality were not largely present. Thus many of our binary variables were imbalanced to largely not be present with some having a present frequency of less than 100 out of 1700. For the continuous variables such as presence of sodium and potassium in the blood, we used frequency bar plots and they largely resembled normal distributions, which also made sense with the data as the more extreme values were unlikely to be common. After this initial visualization, the next step was to create correlation plots to find which categories are the most correlated.



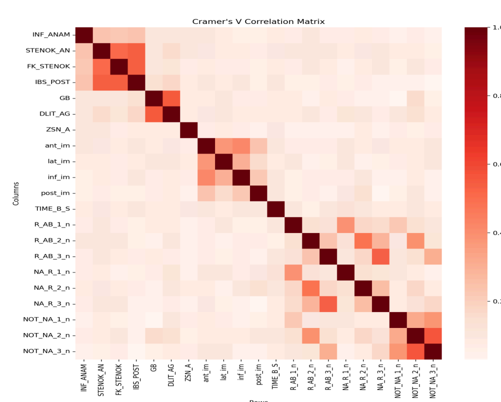
We wanted to create a single plot at first that would find the correlation between all the variables, but we found this to be too difficult due to the different data types as there were not any correlation coefficients which would do what we wanted. We then thought about changing all the variables into continuous variables, so that we could use the Pearson's Correlation Coefficient, but this idea was not feasible as it would involve us changing every categorical and binary variable into a continuous metric that would not only make sense for the specific variable but then also scale it within the same bounds as the rest of the variables. This process would've been extremely tedious but also may not even have given us an accurate understanding of the dataset due to the amount of data manipulation that would've been needed. Thus we chose to use a different correlation coefficient for each data type and then display these values in their respective heatmaps.

For the binary predictors, we chose the Phi Coefficient Correlation Matrix as it specializes in measuring the strength of the association between two binary variables. We found that the Phi Coefficient is equal to that of a Pearson Coefficient for Dichotomous variables which we are dealing with. The results showed that there was a positive correlation between variables that depend on each other such as



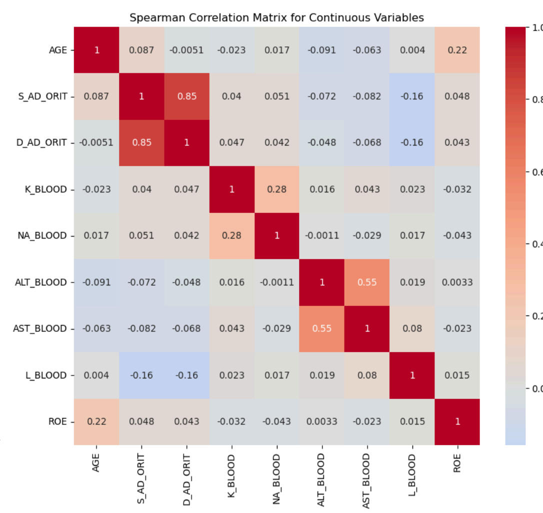
MP_TP_POST (Paroxysms of atrial fibrillation at the time of admission to intensive care unit, or at a pre-hospital stage) and ritm_ecg_p_02 (ECG rhythm at the time of admission to hospital: atrial fibrillation) were positively correlated because both variables indicate the presence of atrial fibrillation at different stages of patient admission, highlighting that patients with pre-hospital or ICU paroxysms of atrial fibrillation are likely to present with atrial fibrillation on hospital admission ECG and negative correlation in variables that are mutually exclusive such as ritm_ecg_p_01 and ritm_ecg_p_07, which binary features for varying ranges of heart rates once given certain medication. Overall, the matrix had many areas with low or no correlation, indicating that many binary variables are relatively independent of each other.

For Categorical variables, we decided to use Cramér's V because it is the correlation of two variables and behaves similarly to that of the Phi correlation Coefficient, except it can be used on variables which have more than 2 levels. Cramér's V has a range of 0 to 1, and can reach 1 only when each variable is completely determined by the other. We found that Cramér's V can be a heavily biased estimator of its population which can lead to large overestimation of strength of association, thus we used a bias correction in our function in our output of the correlation heatmap. For this we found evidence that the three categorical variables that had the strongest correlation were that of IBS_POST (history of



coronary heart disease), STENOK_AN (history of exertional angina pectoris), and FK_STENOK (functional class of angina pectoris) as they are all related as they describe different aspects of coronary artery disease. These variables collectively reflect the occurrence and severity of heart-related events and symptoms, indicating the overall burden and impact of ischemic heart disease on a patient's health. Another interesting trend we found is that the categories which indicate use of opioids were somewhat correlated to the relapse of pain on the same day. The smallest being opioid use on day 1 and pain relapse on day 1 and the largest being use of opioids on day 3 with a relapse of pain showing on day 3 since being admitted into the hospital. These values were still low, around 0.3-0.5 range, but were much larger than the majority of the other correlations between categorical data types.

Lastly, we used the Pearson's Correlation Coefficient for the continuous variables. The reason we opted for Pearson's was because it would determine whether there was a linear relationship between two continuous categories, but also can measure whether a relationship is significant or not. It is the ratio between the covariance of two variables and the product of their standard deviations, so the result will always be between -1 and 1. The only downside is that it ignores all types of other relationships that are not linear that may be present in a dataset like this, however we were more interested in the strength of the relationship directly between two variables for this part of the analysis. With this visualization, we found that D_AD_ORIT (Diastolic blood pressure in the intensive care unit) and S_AD_ORIT (Systolic blood pressure in the intensive care unit) have a strong positive correlation (0.85) because both measurements reflect blood pressure levels, which are inherently related as part of the cardiovascular system's function. When systolic blood pressure is high, diastolic blood pressure typically also tends to be high, and vice versa. We also found that ALT_BLOOD (Serum AlAT content) and AST_BLOOD (Serum AsAT content) have a moderate positive correlation (0.55), suggesting that higher levels of alanine aminotransferase (ALT) are associated with higher levels of aspartate aminotransferase (AST). Both enzymes are markers of liver function, and their levels often rise together in response to liver damage or inflammation. Lastly, we found that to an extent, the amount of Potassium found in the blood, (K_BLOOD) is positively correlated to the amount of Sodium (NA_BLOOD) found in the blood at 0.28, meaning that patients which had presence of one higher than usual would also have the other higher too.

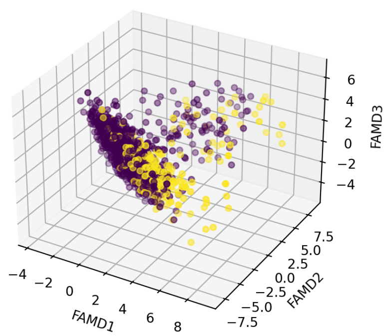


Overall, we found that there was not a significant amount of correlation between the variables within each data type, and this is may because there was correlation that we couldn't explore across data types. However, with this information, we learnt more about the dataset and how these variables interact with each other, which meant that we could move into our statistical analysis portion of the project,

3. Methodology

3.1. Dimensionality Reduction

Given the high dimensionality of our dataset, we chose to implement Factor Analysis of Mixed Data (FAMD). FAMD is a dimensionality reduction technique that can be applied to quantitative and qualitative data. The algorithm standardizes quantitative data, and converts all qualitative data into columns of binary indicators. Singular Value Decomposition is then performed on the data matrix, and the obtained principal components are the directions in space along which the data have the most variance.



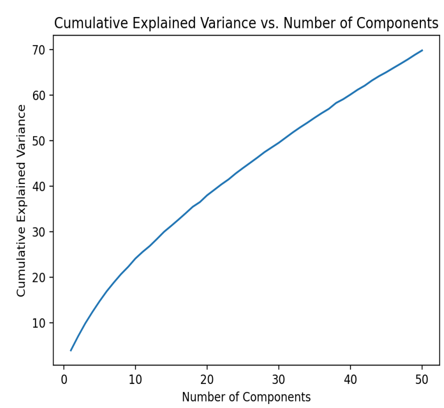
We first implemented FAMD with three components to visualize underlying structure within the data. We plotted the principal components, and labeled the data points by class. The graph displays the output of the analysis.

For clarity, the yellow points denote Class 1 (Lethal Outcome), and the purple points denote Class 0 (Patient Survival). We are able to identify some separation between the classes - specifically noticing that patients who passed away typically had higher values of the first component.

It is also important to note that the components themselves are interpretable. We found that FAMD1 (the first component) was strongly influenced by heart rate and chest pain measurements. FAMD2 was strongly influenced solely by heart rate measurements, and FAMD3 was strongly influenced by blood-electrolyte levels. This interpretation can be tied back to our earlier visual analysis. We noticed that high FAMD1 scores were common among Class 1 patients, which implies that Class 1 patients had abnormal heart rate and chest pain measurements. This is an intuitive conclusion to draw, as chest pain and atrial fibrillation (high heart rate) are often precursors to cardiovascular fatalities.

3.2. Statistical Learning

To develop a method that would effectively identify a patient's outcome (survival/death), we chose to leverage statistical learning. After careful consideration of the data and our objective, we implemented a Gradient Boosted Model. This model's robust flexibility and interpretability allowed us to create an accurate predictor, while also allowing us to identify key factors that drove the predictions. The specific library we used for this task was XGBoost, which is a package that contains Gradient Boosted Models.



However, before fitting the model, we had to identify the data that would be used in the learning process. For the sake of interpretability, we decided that it was sensible to shrink the dataset. This would allow us to analyze the impact of each component on predictions, and ease the computational load. As mentioned before, we performed FAMD with three components to visualize the underlying structure of the data. However, by reducing the dataset to three dimensions, we found that more than 90% of the explained variance in the data was lost. This would intuitively result in poor predictive performance. Therefore, to identify an optimal size for our dimensionality reduction, we performed FAMD with various components, and plotted the cumulative explained variance against the number of components. From the plot, we noticed that the tradeoff between dimensionality and information loss was mostly linear. However, when ten components were used, there seemed to be an "elbow" in the curve. This suggested that using ten components in the reduction was an appropriate choice for balancing the information-dimensionality tradeoff.

After performing the reduction, we were able to interpret the columns. To the left is a description of the type of data that influenced each component the most. We can see that the new components contain similar information to the original three dimensions. Heart rate and chest pain still seem to be the most strong indicators of a patient's outcome.

FAMD1	High Heart Rate + Chest Pain
FAMD2	High Heart Rate
FAMD3	Blood-Electrolyte Levels
FAMD4	Blood Pressure
FAMD5	Presence of Myocardial Infarction
FAMD6	Heart Problems in Medical History
FAMD7	Abnormal ECG Measurements
FAMD8	Chest Pain
FAMD9	Abnormal ECG Measurements
FAMD10	Irregular Heartbeat + Chest Pain

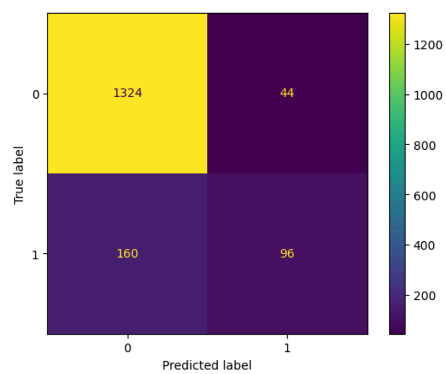
With the dimensionality reduction complete, we were able to proceed with the model training process. To find the best estimator using XGBoost, we implemented 10-Fold Cross-Validation. This process allowed us to train and test the model on multiple folds of the data, and identify the optimal parameters. We found that the best model had 400 estimators, a learning rate of 0.07, and maximum depth of 1.

After finding the best estimator, we measured its performance using 10-Fold Stratified Cross-Validation. The need for stratification stemmed from the class imbalance in the dataset: approximately 81% of the data corresponded to Class 0, and only 19% corresponded to Class 1. By using a stratified split, we maintained this proportion in every test fold, to ensure that the estimator wasn't being trained and tested on misrepresentative data.

4. Results

4.1. Model Performance

The model evaluation process yielded interesting results. From the confusion matrix, we can see that the model predicted Class 0 very well. However, it performed poorly when predicting Class 1. Part of this issue can be attributed to the class imbalance - the model most likely struggled to correctly identify lethal outcomes due to the lack of data.



The classification report reiterates this conclusion. We see that Precision and Recall are high for Class 0. Both of these scores are lower for Class 1, but the Recall is especially low. This indicates that the model tends to label Class 1 observations as Class 0.

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              precision    recall  f1-score   support

           0       0.89      0.98      0.93      1368
           1       0.69      0.38      0.48       256

    accuracy                           0.87      1624
   macro avg       0.79      0.67      0.71      1624
weighted avg       0.86      0.87      0.86      1624
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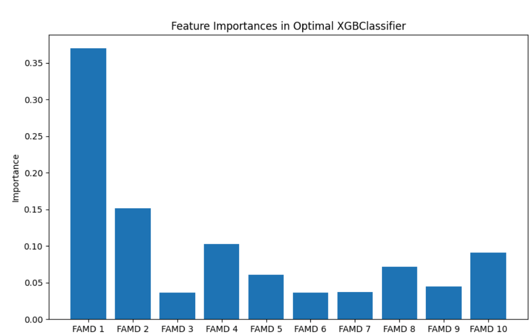
4.2. Interpretation



Using the feature importance attribute of tree-based methods, we were able to identify the variables that had the strongest influence on the predictions. The feature importance plot shows us that the first FAMD component had an overwhelmingly large impact, while the second and fourth components also played important roles. When looking back at the component analysis in the previous section, we see that FAMD1 corresponded to high heart rate and chest pain, FAMD2 was influenced solely by high heart rate, and FAMD4 was a product of blood pressure-related data. We can therefore conclude from our analysis that the most important precursors in heart-related deaths are Atrial Fibrillation (high heart rate), Angina Pectoris (chest pain), and abnormal blood pressure readings.

5. Conclusion and Reflection

5.1. Reflection

Our project demonstrated several strengths in predicting mortality risk in patients with myocardial infarction. Notably, the application of Factor Analysis of Mixed Data (FAMD) effectively managed the dataset's high dimensionality, isolating key predictors such as heart rate, chest pain, and blood pressure measurements that strongly influenced patient outcomes. The use of XGBoost as a predictive model proved robust and interpretable, further validated through extensive cross-validation techniques, ensuring reliable predictions for surviving patients.

Despite these strengths, our model exhibited significant weaknesses. A critical issue was the minimal analysis of individual features, which may have restricted our understanding of the specific impact of each predictor. Furthermore, the class imbalance in our dataset, with a majority of data corresponding to patient survival, resulted in poor predictive performance for lethal outcomes. This imbalance likely caused the model to struggle in accurately identifying lethal outcomes, as evidenced by the low recall for Class 1 predictions.

Future improvements are necessary to address these limitations. Enhancing pattern analysis and data visualization techniques could provide deeper insights into the features, allowing for a more detailed understanding of predictors. Additionally, testing more methods to account for temporal influence could improve the model's accuracy over time. Acquiring more data to balance the classes is crucial, as a more balanced dataset would enhance the model's ability to predict lethal outcomes accurately. Leveraging findings from correlation analyses for feature selection could refine the model further, ensuring that only the most relevant predictors are utilized, thereby improving overall performance.

5.1. Conclusion

Our project demonstrated the feasibility and potential of using data science techniques to predict mortality risk in patients with myocardial infarction. The key predictors identified included heart rate, chest pain levels, and abnormal blood pressure readings. However, the model's performance was hampered by class imbalance, highlighting the need for more balanced datasets and additional techniques to address this issue. By continuing to refine and expand upon our methodologies, we can contribute to more effective and targeted interventions, ultimately improving patient outcomes and reducing mortality rates. The integration of advanced pattern analysis, data visualization, and comprehensive feature analysis will be instrumental in advancing our predictive capabilities and achieving more reliable results in future applications.